WORKED SOLUTION

STEP II 2000, Question 7

Pure • Lines and Cones

STEP II 2000, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q7

The key idea

Before doing anything else, work out the length of $\mathbf{s}$. The $\cos\theta \pm \sqrt{3}$ terms are set up so that the cross terms cancel and $\cos^2\theta + \sin^2\theta$ appears, so $|\mathbf{s}|$ doesn't depend on θ at all. After that, everything comes from the scalar product. The combination $s_x - s_z$ is also free of θ, which points straight at the direction of m and explains why the plane $x - z =$ constant appears in the final part.

Solution

The angle between l and a general line

Write $c_\theta = \cos\theta$ for short. The length of $\mathbf{s}$ is

$$\begin{aligned} |\mathbf{s}|^2 &= (c_\theta + \sqrt{3})^2 + 2\sin^2\theta + (c_\theta - \sqrt{3})^2 \\ &= 2c_\theta^2 + 6 + 2\sin^2\theta = 8, \end{aligned}$$

since the $\pm 2\sqrt{3}\,c_\theta$ cross terms cancel. So $|\mathbf{s}| = 2\sqrt{2}$ for every θ.

Let $\mathbf{d} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$. Then

$$\mathbf{s}\cdot\mathbf{d} = a(c_\theta + \sqrt{3}) + \sqrt{2}\,b\sin\theta + c(c_\theta - \sqrt{3}) = (a+c)\cos\theta + \sqrt{2}\,b\sin\theta + \sqrt{3}\,(a-c).$$

So the angle ϕ between the two lines satisfies

$$\cos\phi = \frac{(a+c)\cos\theta + \sqrt{2}\,b\sin\theta + \sqrt{3}\,(a-c)}{2\sqrt{2}\,\sqrt{a^2+b^2+c^2}}.$$

A line has no preferred direction, so if you want the acute angle, take the modulus of the right-hand side.

The line m

We want a direction $\mathbf{d}$ that makes $\cos\phi$ independent of θ. Looking at the numerator, the θ-dependence goes away exactly when $a + c = 0$ and $b = 0$. So try $\mathbf{d} = \mathbf{i} - \mathbf{k}$ ($a = 1$,

$b = 0$, $c = -1$):

$$\cos\phi = \frac{\sqrt{3}\cdot 2}{2\sqrt{2}\cdot\sqrt{2}} = \frac{\sqrt{3}}{2}.$$

This is positive, so ϕ is already acute, and $\phi = \pi/6$ for every θ. Hence the line

$$m: \quad \mathbf{r} = \mu(\mathbf{i} - \mathbf{k})$$

through the origin does what is required. As θ varies, l sweeps out a cone with axis m and semi-vertical angle $\pi/6$.

The circle traced by P

On l, $x - z = \lambda(s_x - s_z) = \lambda((c_\theta + \sqrt{3}) - (c_\theta - \sqrt{3})) = 2\sqrt{3}\,\lambda$. Setting this equal to $4\sqrt{3}$ gives $\lambda = 2$ whatever θ is, so

$$\overrightarrow{OP} = 2\mathbf{s} = (2\cos\theta + 2\sqrt{3})\,\mathbf{i} + 2\sqrt{2}\sin\theta\,\mathbf{j} + (2\cos\theta - 2\sqrt{3})\,\mathbf{k}.$$

The line m meets the plane where $\mu - (-\mu) = 4\sqrt{3}$, that is $\mu = 2\sqrt{3}$, at the point C with

$$\overrightarrow{OC} = 2\sqrt{3}\,\mathbf{i} - 2\sqrt{3}\,\mathbf{k}.$$

? **Ask yourself** Why is C the natural candidate for the centre?

The normal to the plane $x - z = 4\sqrt{3}$ is $\mathbf{i} - \mathbf{k}$, which is the direction of m. So m is the axis of the cone, it is perpendicular to the plane, and any cone cut by a plane perpendicular to its axis gives a circle centred where the axis crosses that plane. Now we check this directly:

$$\overrightarrow{CP} = 2\cos\theta\,\mathbf{i} + 2\sqrt{2}\sin\theta\,\mathbf{j} + 2\cos\theta\,\mathbf{k}, \qquad |\overrightarrow{CP}|^2 = 4\cos^2\theta + 8\sin^2\theta + 4\cos^2\theta = 8.$$

So every position of P lies in the plane and at distance $2\sqrt{2}$ from the fixed point C of that plane. In other words, P lies on the circle in the plane with centre C and radius $2\sqrt{2}$, and C lies on m.

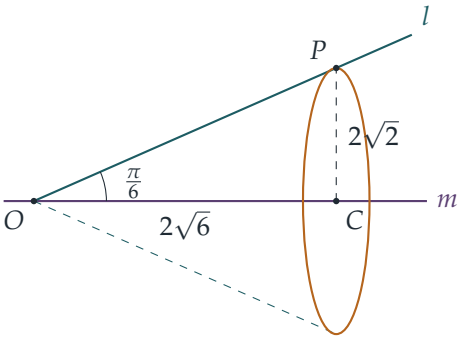
To see that P really traces the whole circle, write

$$\overrightarrow{CP} = 2\sqrt{2}\,(\cos\theta\,\mathbf{u} + \sin\theta\,\mathbf{v}), \qquad \mathbf{u} = \tfrac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{k}), \quad \mathbf{v} = \mathbf{j}.$$

Here $\mathbf{u}$ and $\mathbf{v}$ are perpendicular unit vectors, both perpendicular to $\mathbf{i} - \mathbf{k}$, so they lie along the plane. As θ runs from 0 to 2π, P goes once round the circle. The radius is

$$\boxed{2\sqrt{2}}.$$

A quicker way to get the radius: $|OP| = 2|\mathbf{s}| = 4\sqrt{2}$ and $|OC| = 2\sqrt{6}$, and OC is perpendicular to the plane, so Pythagoras gives $CP = \sqrt{32 - 24} = 2\sqrt{2}$. Equivalently, $CP = OC\tan\frac{\pi}{6} = 2\sqrt{6}/\sqrt{3}$.



Checks

- At $\theta = 0$, $\mathbf{s} = (1+\sqrt{3}, 0, 1-\sqrt{3})$, with $|\mathbf{s}|^2 = (4+2\sqrt{3}) + (4-2\sqrt{3}) = 8$, and $\mathbf{s}\cdot(\mathbf{i}-\mathbf{k}) = 2\sqrt{3}$, giving $\cos\phi = \sqrt{3}/2$.
- The two routes to the radius agree: $\sqrt{(4\sqrt{2})^2 - (2\sqrt{6})^2} = \sqrt{8}$ and $2\sqrt{6}\tan\frac{\pi}{6} = \sqrt{8} \approx 2.828$.
- Numerically, for 20 000 random θ (and random a, b, c for the general formula), the angle with m equals $\pi/6$, P lies in the plane, and $|CP| = 2.8284\ldots$, all to within 10^{-15}.

Common slips

- Forgetting to divide by $|\mathbf{s}|$, or assuming $\mathbf{s}$ is a unit vector. It has length $2\sqrt{2}$.
- Showing only that P lies on a sphere about O (that is, $|OP| = 4\sqrt{2}$). A sphere cut by a plane gives a circle, but you still have to say why its centre is C, on m: it is because OC is perpendicular to the plane.

Beyond STEP

The line l sweeps out a circular cone, and cutting a cone with a plane is how the conic sections got their name. A plane perpendicular to the axis gives a circle; tilt it and you get an ellipse, a parabola or a hyperbola. In a first-year course you would describe the cone as the quadric surface $(\mathbf{r}\cdot\hat{\mathbf{n}})^2 = |\mathbf{r}|^2\cos^2\alpha$ and classify such surfaces using eigenvalues of a symmetric matrix. Strang's *Introduction to Linear Algebra* covers this.

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